

Bike Sharing Demand Prediction

Heike Thöricht | Bremen | 23.08.2026 | GitHub: https://github.com/Thoericht/bike_sharing

Objective

The objective of this project is to predict the hourly number of bike rentals (cnt) using the UCI Bike Sharing Dataset. The analysis includes exploratory data analysis (EDA), feature engineering and model comparison to identify a model suitable for a production forecasting service.

Exploratory Data Analysis

The dataset contains **17,379 hourly observations** from 2011–2012 with no missing values.

The EDA identified three main findings:

- Bike rental demand strongly depends on the **hour of the day**, with pronounced morning and evening peaks on working days.
- **Temperature** positively influences demand, while adverse weather conditions reduce rentals.
- The target variable is **right-skewed**, with relatively few hours showing very high demand.

Based on these findings, two additional features were evaluated: time_of_day and temp_diff. A feature derived from registered users was excluded because it caused **target leakage**.

Model Comparison

Three models were evaluated using a time-based train-test split and TimeSeriesSplit cross-validation.

Model	R ²	MAE
Random Forest (Baseline)	0.862	54.57
Random Forest (Engineered)	0.880	50.39
Linear Regression	0.632	98.87
XGBoost	0.897	47.35

Selected Model

XGBoost was selected because it achieved the best predictive performance on both the hold-out test set and the time-series cross-validation. It models non-linear relationships effectively and handles interactions between temporal and weather-related features well.

The grouped feature importance shows that **time-related variables (hr and time_of_day)** are the **dominant predictors**, which is consistent with the EDA.

Final performance:

- **R²**: 0.897
- **MAE**: 47.35 rentals/hour
- **RMSE**: 70.68

Production Considerations

The implementation uses reusable scikit-learn pipelines, consistent preprocessing, time-based validation and modular evaluation functions to support maintainability and reproducibility. For a production environment, the project could be extended with automated testing, hyperparameter tuning and model monitoring.

